Challenges

There are multiple challenges teams can face in implementing and deploying AI based virtual assistants.

Cost could be a major factor. There are many commercial tools available in the market, but they are expensive. Additionally, we need to buy plugins to interface with enterprise systems. One commercial tool may not be sufficient. For small- and medium-sized companies this cost can be a challenge. Open source modules, and customised orchestration engines suitable for specific enterprise needs, can help reduce these costs. Table 3 lists the recommended open source modules for on-premises deployments. In case of public cloud deployments, we can consider utilising native SaaS services from the cloud provider.

Integrating virtual assistants into your existing systems and applications can be a challenge as some level of in-house development effort is required. Home-grown scripts need to be maintained and updated as and when systems change. The return on investment of these efforts is high and justifiable.

Overcoming resistance can be another challenge. Team members may think AI can pose a risk to their jobs. But AI is all about relieving employees of repetitive work and freeing them up to be more creative and perform to their real potential.

The reasons for each IT service failure can be unique and root cause analysis might yield unexpected results. AI assistants help to analyse data systematically, learn from earlier failures and provide a quick resolution. They help in accelerating root cause analysis and restoring IT services that fail. In addition, these assistants help to fast-track IT admin operations with less manual intervention. Small companies can use open source platforms to build AI digital assistants. END

Table 3

Function	Open source modules	Cloud native deployments
Chatbot, NLP engine	ChatSDK, ChatterBot, NTLK	Azure BOT Service, Amazon Lex, Alexa
Automation	Ansible, Python, PowerShell scripts	Serverless functions
Visualisation	D3.js, Grafana	Power BI
Tracing, Dependency analysis	Jaeger	Azure Monitor, AWS CloudWatch
Telemetry	Prometheus, Istio	Azure Monitor, AWS CloudWatch
Security	Python libraries	AWS Security Hub, Azure Sentinel, Security Center
Infrastructure automation	Ansible, Terraform, Jenkins, Chef, Selenium	ARM, Cloud Foundation
AIOPs	Seldon, Logilizer, Aiopstools, Whylogs	Vendor SaaS services

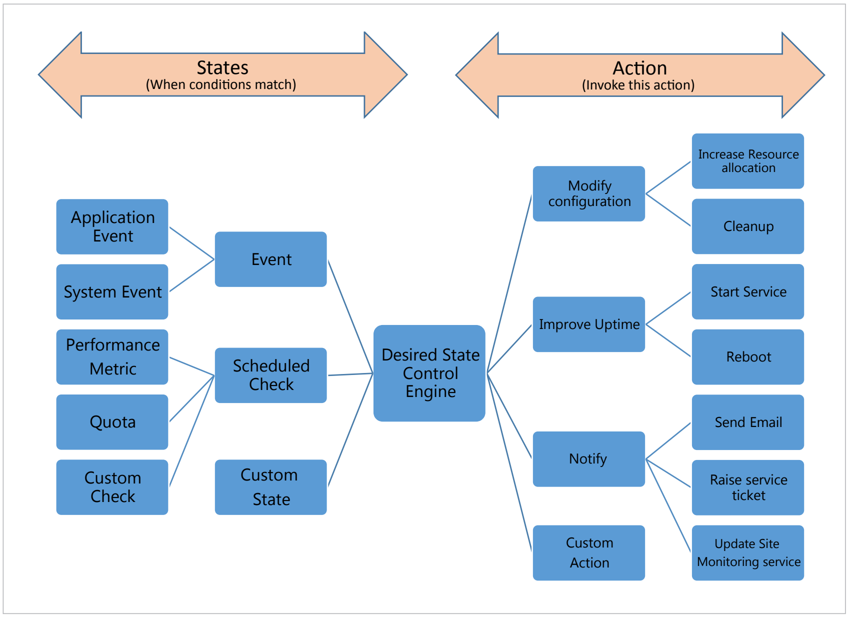

Figure 3: Logical design of an auto-heal assistant

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